

Experiment - 1

AIM: To understand DevOps, principles, practises and DevOps roles and responsibilities.

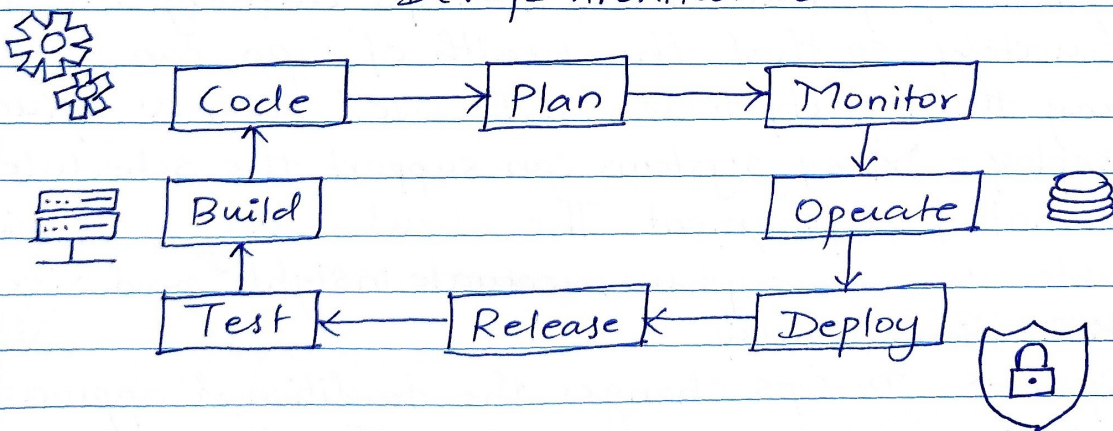
THEORY :

Definition

- * DevOps is the combination of two words, one is Development and other is Operations. It is a culture to promote the development and operation process collectively.
- * DevOps helps to increase organisation speed to deliver applications and services. It also allows organisations to serve their customers better and compete more strongly in the market.
- * DevOps can also be defined as a sequence of development and IT operations.
- * DevOps has become one of the most valuable business disciplines for enterprises. With its help, quality and speed of application delivery has improved.

Architecture

DevOps Architecture



1. Build - Without DevOps, the cost of consumption of the resources was evaluated based on the pre-defined individual usage with fixed hardware allocation. The usage of cloud, sharing of resources comes into picture, which is a mechanism to control the usage of resources or capacity.
2. Code - Many good practises such as Git enables the code to be used, which ensures writing the code for business, helps track changes and if necessary reverting to the original code developed. The code can be appropriately arranged in files, folders etc.
3. Test - The application will be ready for production after testing, it consumes more time in testing and moving the code to output. The testing can be automated, which decreases the time for testing so that the time to deploy code can be reduced.
4. Plan - DevOps uses agile methodology to plan development. With the operations and development team in sync, it helps organising the work.
5. Monitor - Continuous monitoring is used to identify any risk of failure. Also, it helps tracking the system accurately so that the health of app can be check. Many third party tools can be used such as Splunk.
6. Deploy - Many systems can support the scheduler for automated deployment. The cloud management platform enables users to capture accurate insights and view the optimization scenario.
7. Operate - DevOps changes the traditional approach of developing and testing separately. The teams operate in a collaborative way where both the teams actively participate throughout the service lifecycle.

8. Release - Deployment to an environment can be done by automation. But when the deployment is made to the production environment, it is done by manual triggering. Many processes involved in release management commonly used to do deployment in production environment manually to lessen the impact on the customers.

Principles

- * Collaboration
- * Data-based decision making
- * Customer-centric decision making
- * Responsibility throughout the lifecycle
- * Automation
- * Failure as a learning opportunity

Advantages

- * DevOps is an excellent approach for quick deployment.
- * It responds faster to market changes to improve growth
- * DevOps escalate business profit by decreasing software delivery time and transportation costs
- * DevOps clears the descriptive process, which gives clarity on product development and delivery
- * It improves customer experience and satisfaction
- * It simplifies collaboration and places all tools in clouds.
- * DevOps means collective responsibility, which leads to better team engagement and productivity

Disadvantages

- * DevOps professional or expert's developers are less available
- * Developing with DevOps is expensive

- * Adapting new DevOps technology into industries is hard to manage in a short time
- * Lack of DevOps Knowledge can be a problem in continuous integration of automation projects

Conclusion

Hence, we have known what DevOps is and its advantages and disadvantages